



Inspiring Excellence

Department of Electrical and Electronic Engineering
PROJECT REPORT Submission

Semester: SUMMER 2026

Course Code: EEE101L

Course Title: Electrical Circuits I Laboratory

Section: 03

Project Title: *Design of a Two-Source Voltage Monitoring Circuit*

Group No: 1

Group members:

<i>Sl.</i>	<i>Name</i>	<i>ID</i>
1.	RUHINUL HAQUE BHUIYAN	1000058247
2.	MANAJIR TASBIK MURSHED	1000055271
3.	DIBYENDU ROY DIBYA	1000058422

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1. Assigned Requirements

Our group was assigned two DC sources, $V_1 = 10\text{ V}$ and $V_2 = 6\text{ V}$, and asked to design a purely resistive network that reports which of the two sources are currently switched on, using a single output terminal read by a digital multimeter. Each source can independently be ON or OFF. An OFF source is connected to ground and is never left floating. There are four operating states, S0 through S3.

For the two single-source states, the required no-load output voltages are 2.2 V for S1 (V_1 ON, V_2 OFF) and 1.1 V for S2 (V_1 OFF, V_2 ON), each within $\pm 5\%$. For S3, both sources are ON and the expected output is obtained by Superposition from the two single-source contributions, not by adding the two assigned targets directly.

With a 10 k Ω monitoring load connected at the output, the four output levels must remain distinguishable, with adjacent sorted levels separated by at least 0.50 V. The circuit uses resistors only, values are within the specified 1 k Ω to 10 k Ω range, both supplies share a common ground, neither source exceeds 12 V, and source-branch current stays below 10 mA. The same nominal resistor values are used in PSpice and the hardware circuit.

1.1 Groupwise Parameters

Source 1, V_1	Source 2, V_2	S1 target (no-load)	S2 target (no-load)
10 V	6 V	2.2 V	1.1 V

1.2 Required Operating States

State	Source 1	Source 2
S0	OFF (0 V)	OFF (0 V)
S1	ON (10 V)	OFF (0 V)
S2	OFF (0 V)	ON (6 V)
S3	ON (10 V)	ON (6 V)

1.3 Constraints Summary

- Spec 1: No-load S1 and S2 outputs are within $\pm 5\%$ of 2.2 V and 1.1 V, respectively.
- Spec 2: No-load S3 agrees within $\pm 5\%$ with the superposition sum of the two individual no-load contributions using the final resistor values.
- Spec 3: With the 10 k Ω monitoring load connected, adjacent sorted output levels differ by at least 0.50 V.
- Spec 4: Resistors only; standard values between 1 k Ω and 10 k Ω ; common ground; neither source above 12 V; branch current below 10 mA.
- Spec 5: The same nominal resistor values are used in simulation and hardware.

2. Design Development

2.1 Circuit Topology

The final circuit is a resistive summing-junction network. R_1 connects the V_1 branch to output Node A, R_2 connects the V_2 branch to Node A, and R_3 connects Node A to common ground. Each source branch is manually switched between its supply voltage and ground. Node A is the single output point measured by the DMM.

This topology is useful because when one source is OFF, its branch is grounded — which is electrically the same operation as deactivating an ideal voltage source for Superposition analysis. Therefore the S1 and S2 single-source states are also the two individual source contributions needed for the S3 calculation.

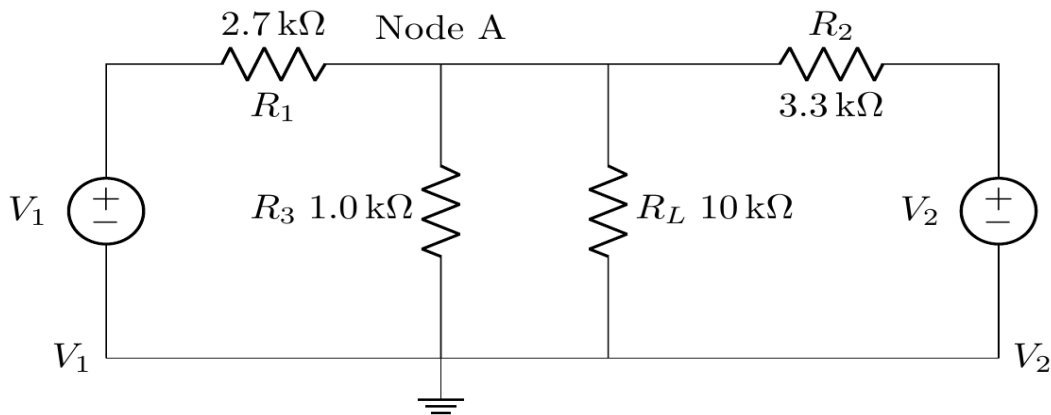


Figure 1: Master schematic. R_L ($10\text{ k}\Omega$) is connected between Node A and ground, in parallel with R_3 , only for loaded-state tests.

2.2 Final Resistor Values

Component	Branch / role	Nominal value
R_1	V1 branch	$2.7\text{ k}\Omega$
R_2	V2 branch	$3.3\text{ k}\Omega$
R_3	Node A to ground	$1.0\text{ k}\Omega$

Standard E12 values within the required $1\text{ k}\Omega$ – $10\text{ k}\Omega$ range, selected to satisfy all five specifications with adequate margin.

2.3 Design Iterations

Our first attempt used only two resistors (R_1 , R_2), meeting directly at the output node with no third resistor to ground. This arrangement reduces to a single voltage-divider ratio for every single-source state, so it could not be tuned to satisfy the S1 and S2 targets independently — fixing the ratio for one target automatically fixed the output for the other, and the two targets do not correspond to the same ratio.

We resolved this by adding R_3 from Node A to ground, which gives the network a second independent ratio to tune. With R_1 , R_2 , and R_3 all adjustable, both single-source targets can be met simultaneously, and standard E12 values were then selected ($R_1 = 2.7\text{ k}\Omega$, $R_2 = 3.3\text{ k}\Omega$, $R_3 = 1.0\text{ k}\Omega$) that satisfy all five specifications with adequate margin.

3. Theoretical Analysis

3.1 Node Definition

Common ground is the reference node at 0 V. The only unknown node voltage is Node A, since the far ends of R1 and R2 are fixed by the source states.

3.2 KCL and Output-Voltage Formula

Using currents leaving Node A and Ohm's Law:

$$\frac{V_A - V_1}{R_1} + \frac{V_A - V_2}{R_2} + \frac{V_A}{R_3} = 0$$

Multiplying through by $R_1 R_2 R_3$ and solving for V_A (entirely in terms of resistor values):

$$V_A = \frac{V_1 R_2 R_3 + V_2 R_1 R_3}{R_1 R_2 + R_1 R_3 + R_2 R_3}$$

3.3 No-Load Calculations

For $R_1 = 2.7 \text{ k}\Omega$, $R_2 = 3.3 \text{ k}\Omega$, $R_3 = 1.0 \text{ k}\Omega$: $R_1 R_2 = 8.91 \text{ k}\Omega^2$; $R_1 R_3 = 2.70 \text{ k}\Omega^2$; $R_2 R_3 = 3.30 \text{ k}\Omega^2$; denominator = $14.91 \text{ k}\Omega^2$.

State	Calculated V_A (no-load)	Target / Reference	% Error
S0	0.000 V	—	—
S1	2.213 V	2.2 V ($\pm 5\%$)	+0.60%
S2	1.087 V	1.1 V ($\pm 5\%$)	-1.23%
S3	3.300 V	See Section 3.4 (superposition)	—

3.4 Superposition Calculation for S3 (Spec 2)

By Superposition, the total response is the sum of the response from V1 alone and V2 alone, with the other source deactivated (deactivating a voltage source = replacing it with a short to ground — exactly what our OFF state already does):

$$V_A(S3) = V'_1 + V'_2$$

$V'_1 = 2.213 \text{ V}$, $V'_2 = 1.087 \text{ V} \rightarrow S3 = 3.300 \text{ V}$.

3.5 Loaded Analysis with $10 \text{ k}\Omega$

The $10 \text{ k}\Omega$ monitoring resistor is connected from Node A to ground, in parallel with R_3 :

$$R'_3 = \frac{R_3 R_L}{R_3 + R_L}$$

$R'_3 = (1.0 \times 10)/(1.0 + 10) = 0.9091 \text{ k}\Omega$. Using R'_3 in place of R_3 in the Section 3.2 formula:

State	Loaded V _A	Sorted (low → high)	Gap to next
S0	0.000 V	S0 = 0.000 V	—
S2	1.025 V	S2 = 1.025 V	1.025 V
S1	2.089 V	S1 = 2.089 V	1.063 V
S3	3.114 V	S3 = 3.114 V	1.025 V

All adjacent gaps clear the 0.50 V minimum required by Spec 3 with more than double the required margin.

3.6 Branch Currents and Power

Branch currents follow from Ohm's Law across each resistor; power follows from $P = I^2R$. The true worst case for each resistor (its single largest current across all 8 state/load combinations) is:

Resistor	Worst-case current	Worst-case power	Requirement
R1 = 2.7 k Ω	2.930 mA (S1, loaded)	23.2 mW	< 250 mW
R2 = 3.3 k Ω	1.508 mA (S2, loaded)	7.5 mW	< 250 mW
R3 = 1.0 k Ω	3.300 mA (S3, no-load)	10.9 mW	< 250 mW

All currents stay below the 10 mA limit (Spec 4) with better than 3 $\times$ margin, and all resistor powers stay below 5% of a standard 1/4 W (250 mW) rating.

4. Simulation

PSpice Schematics was used to simulate all four source states, first without the 10 k Ω monitoring resistor and then with it. OFF sources were represented by setting their DC value to 0 V, which is electrically identical to grounding that branch. Because the circuit is purely resistive with constant DC sources, only a Bias Point (DC operating point) analysis was needed — no transient or AC sweep setup was required. Voltage and current markers were placed at Node A and on the R1/R2 branches to read the results directly off the schematic.

4.1 Simulation Result Table

State	Calculated no-load	PSpice no-load	Calculated loaded	PSpice loaded
S0	0.000 V	0.000 V	0.000 V	0.000 V
S1	2.213 V	2.213 V	2.089 V	2.089 V
S2	1.087 V	1.087 V	1.025 V	1.025 V
S3	3.300 V	3.300 V	3.114 V	3.114 V

Every simulated value matches the calculated value to within PSpice's own display rounding (< 0.001 V), confirming the simulated circuit is wired exactly as designed.

4.2 PSpice Evidence

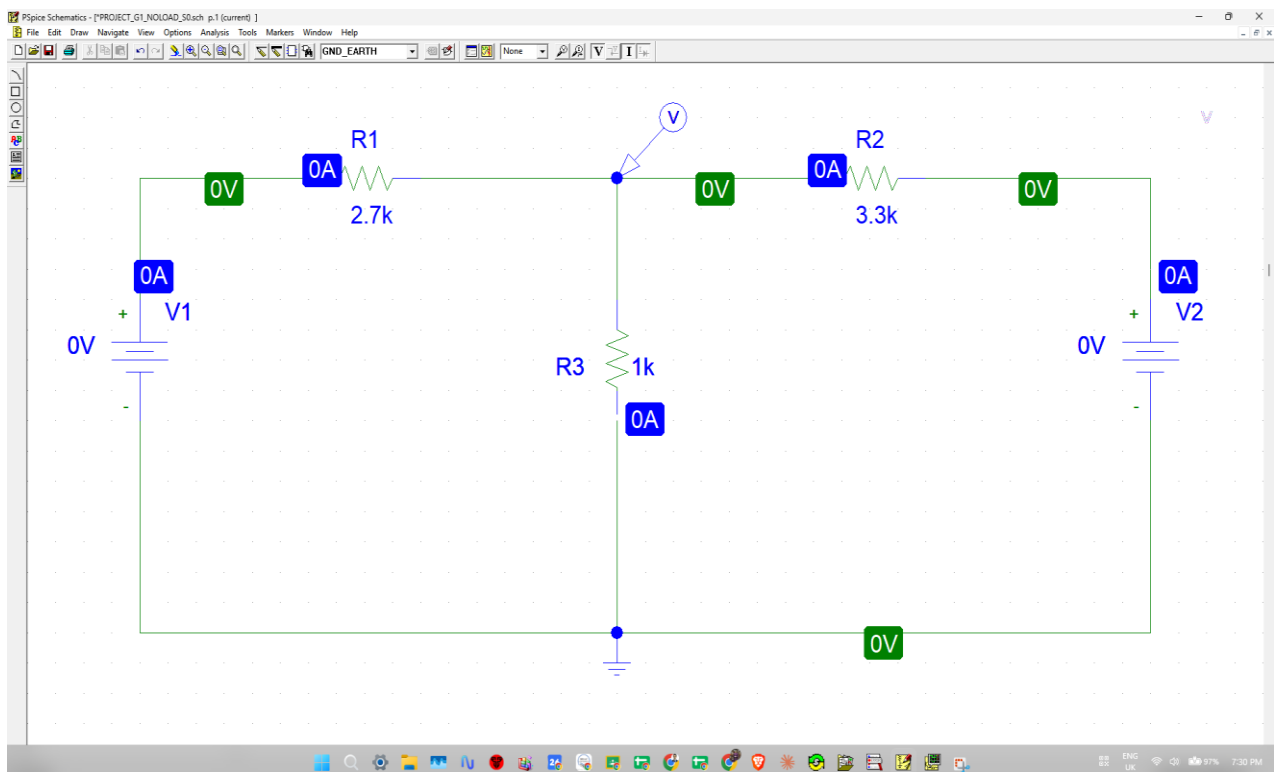


Figure: PSpice simulation for State S0, no-load.

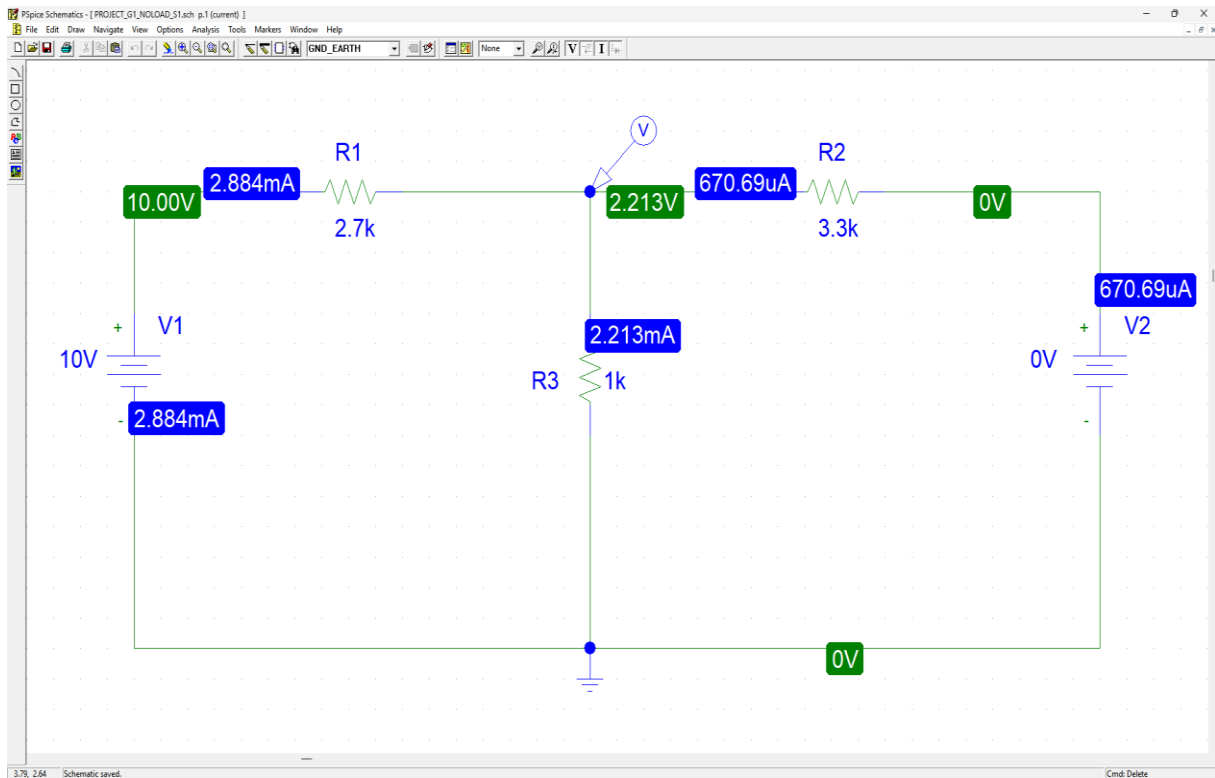


Figure: PSpice simulation for State S1, no-load.

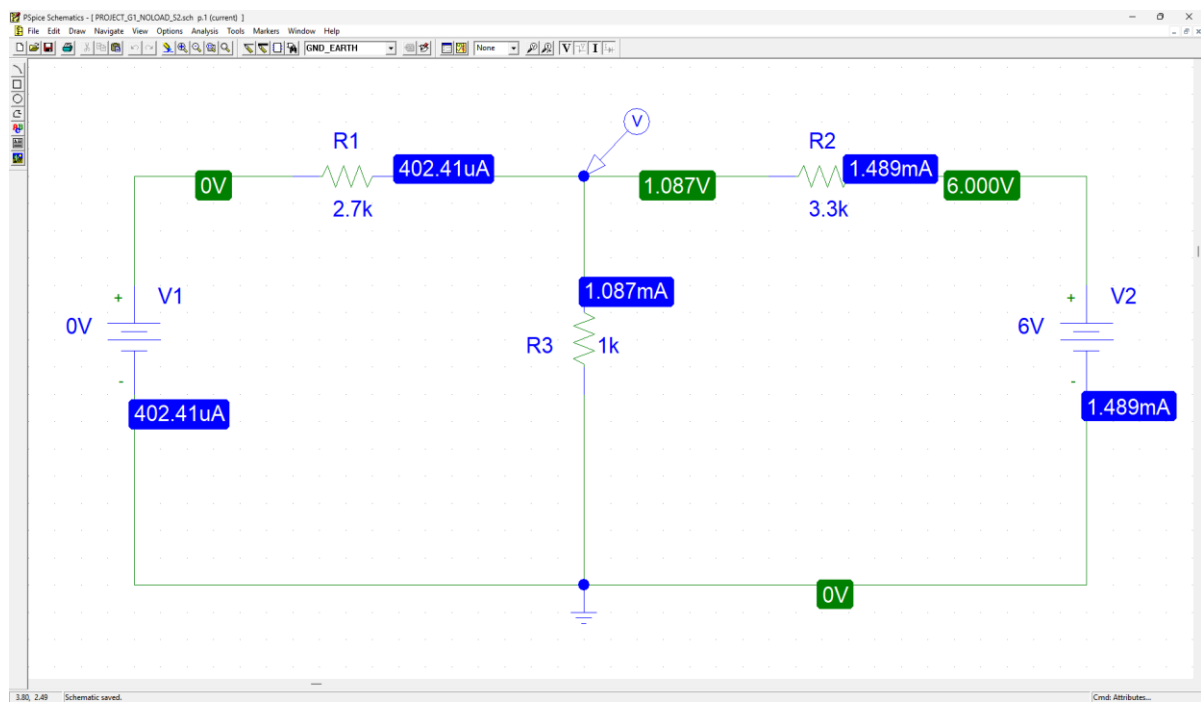


Figure: PSpice simulation for State S2, no-load.

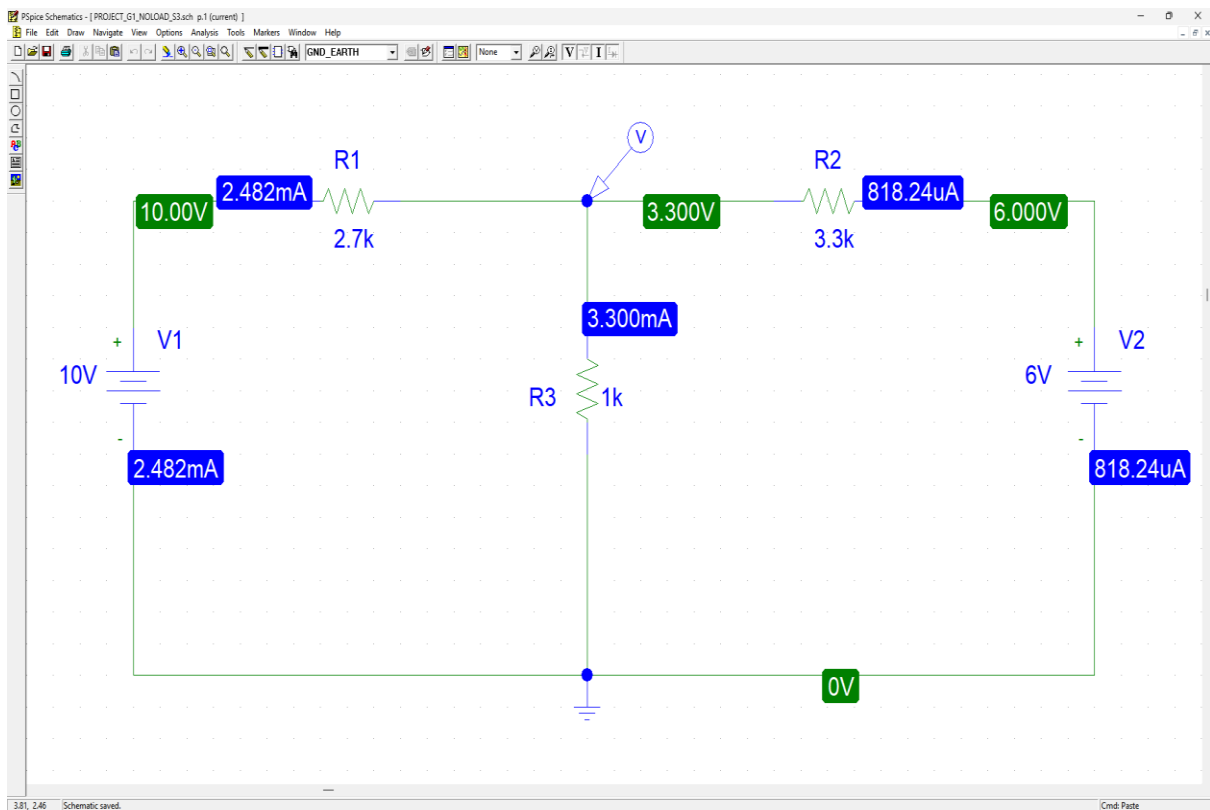


Figure: PSpice simulation for State S3, no-load.

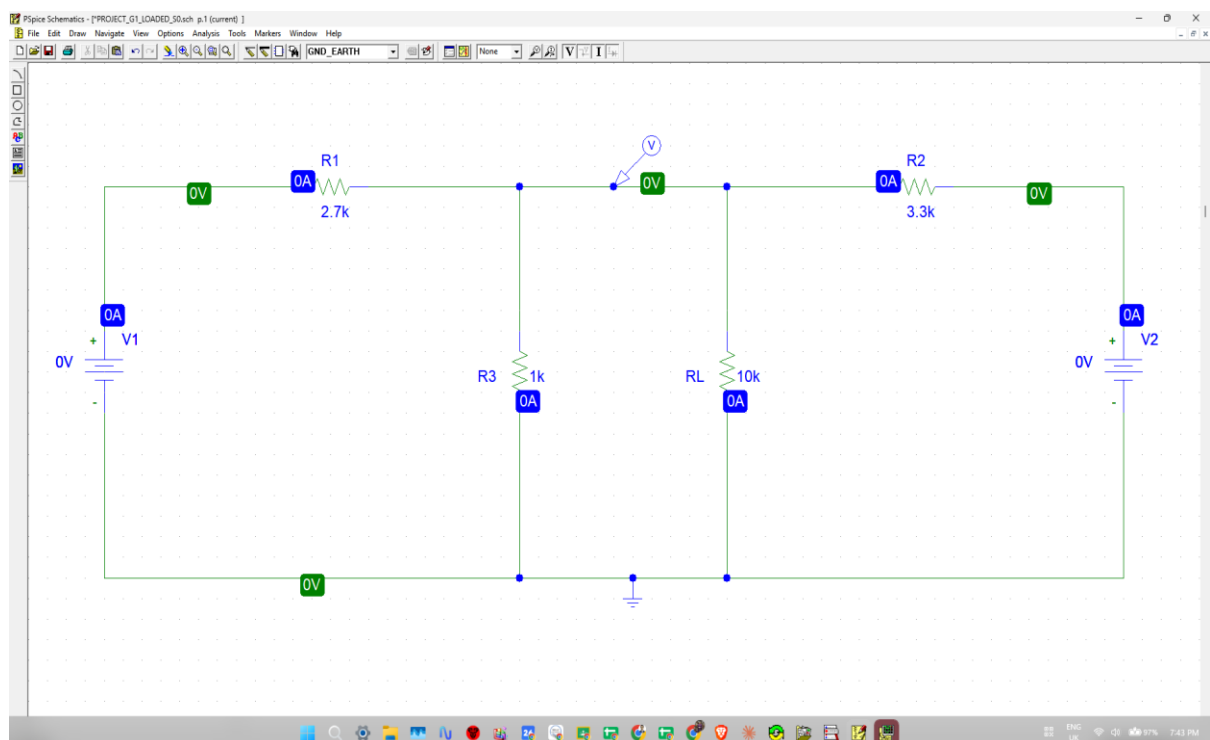


Figure: PSpice simulation for State S0, loaded (10 kΩ).

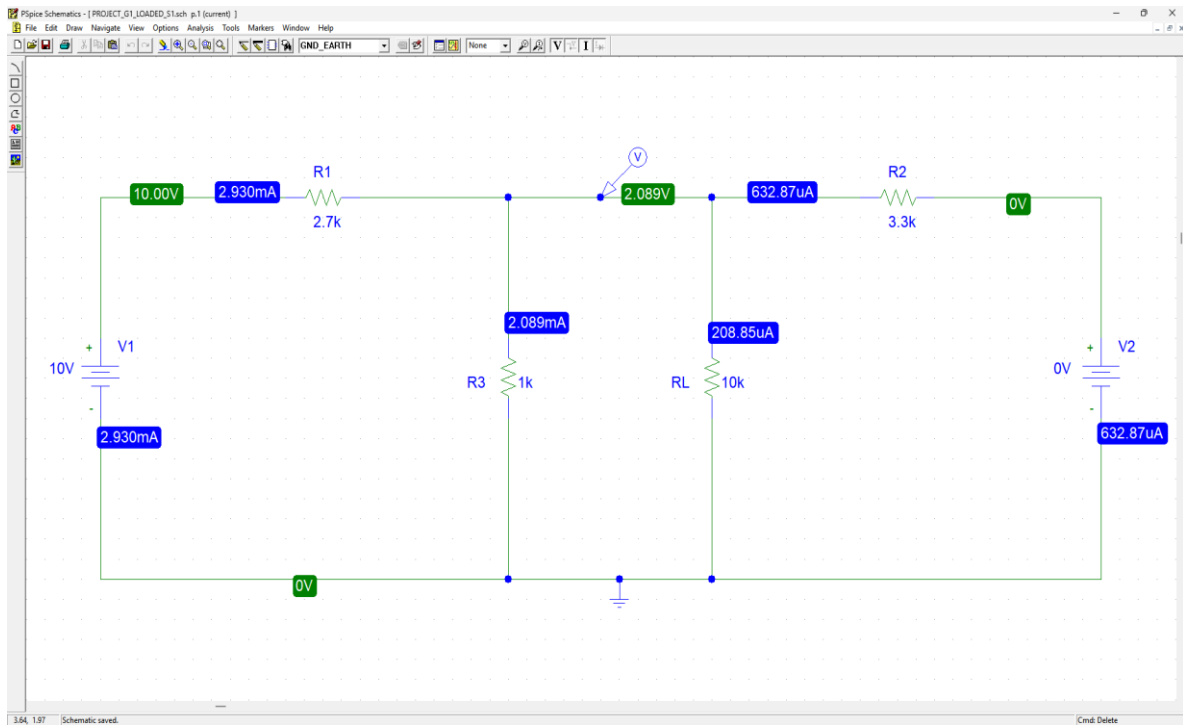


Figure: PSpice simulation for State S1, loaded ($10\text{ k}\Omega$).

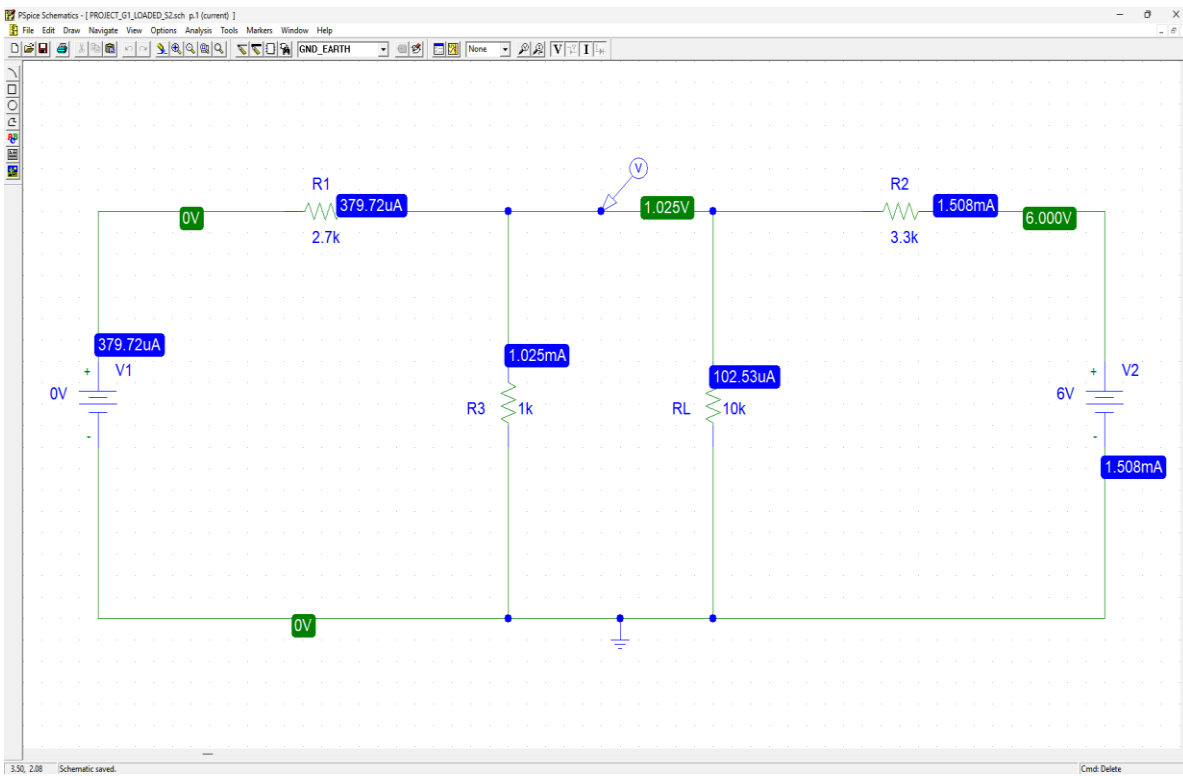


Figure: PSpice simulation for State S2, loaded ($10\text{ k}\Omega$).

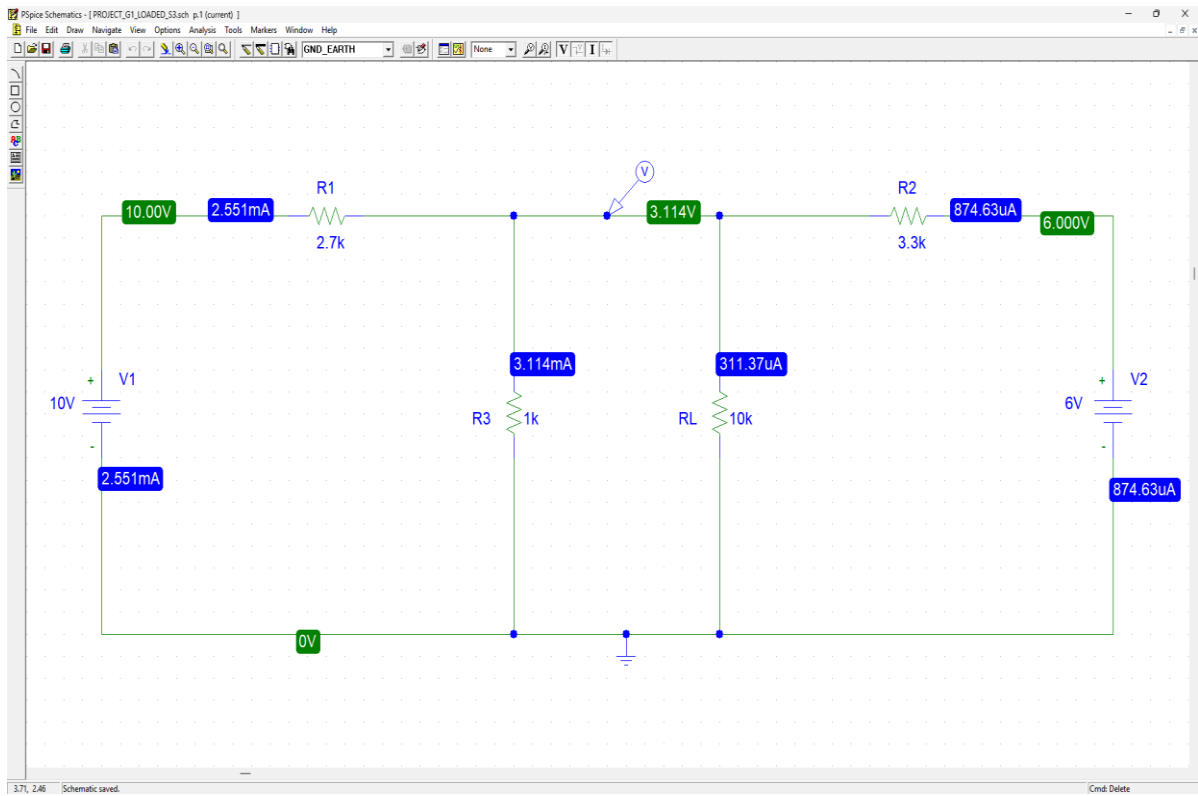


Figure: PSpice simulation for State S3, loaded ($10\text{ k}\Omega$).

5. Hardware Implementation

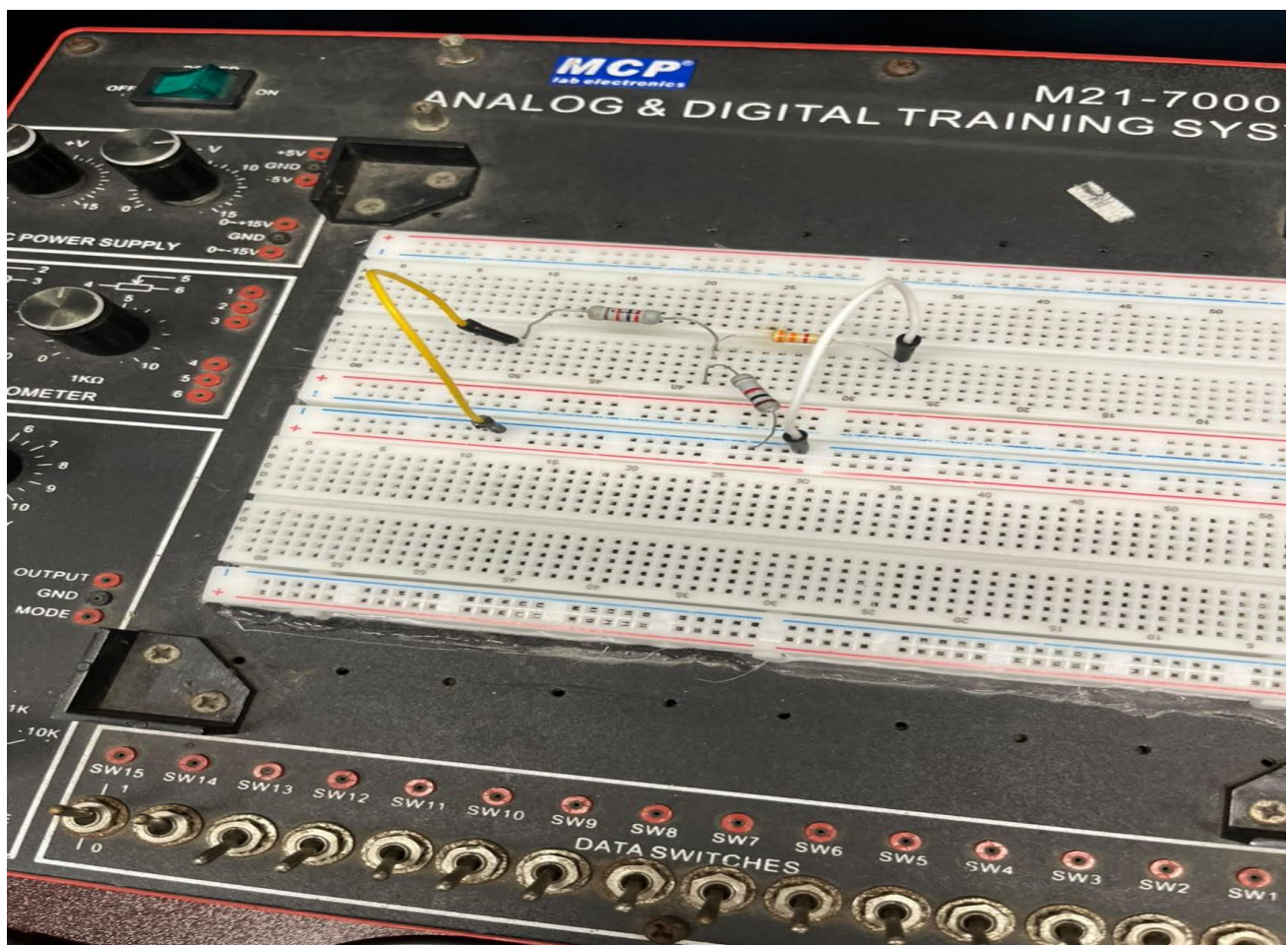
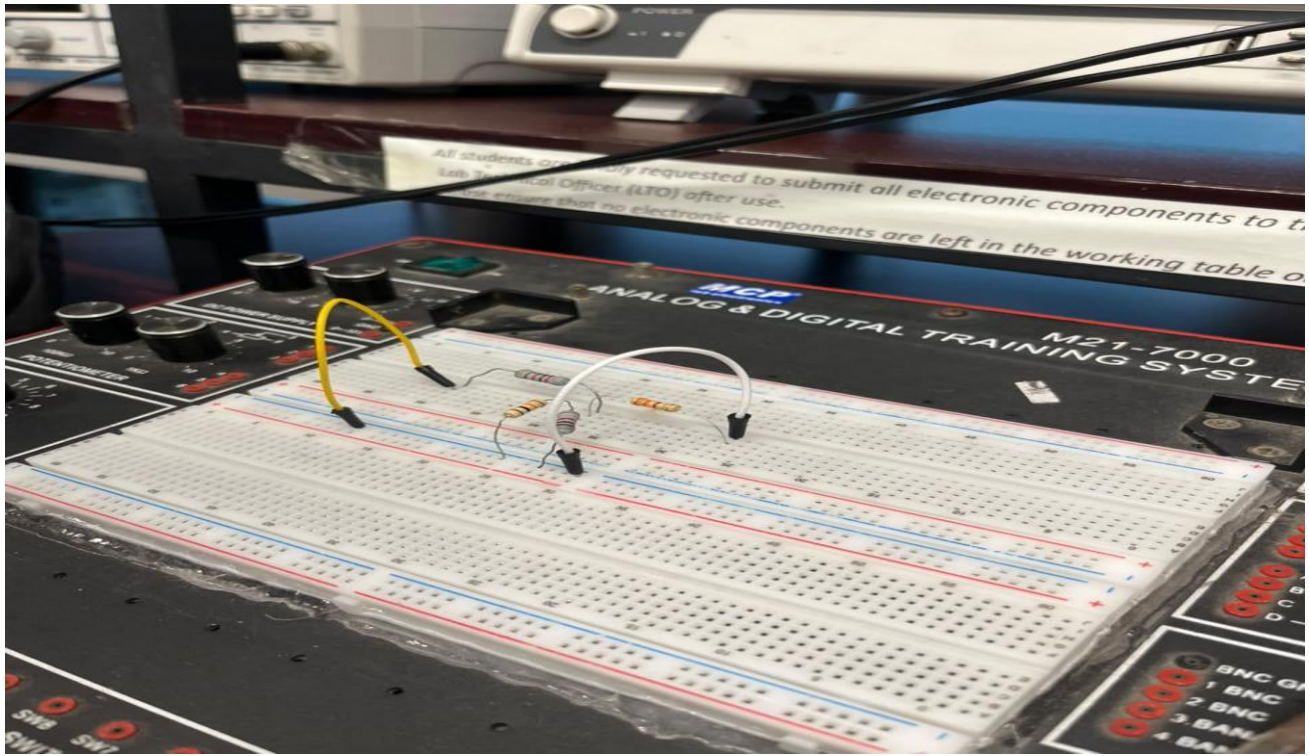
5.1 Actual Component Values

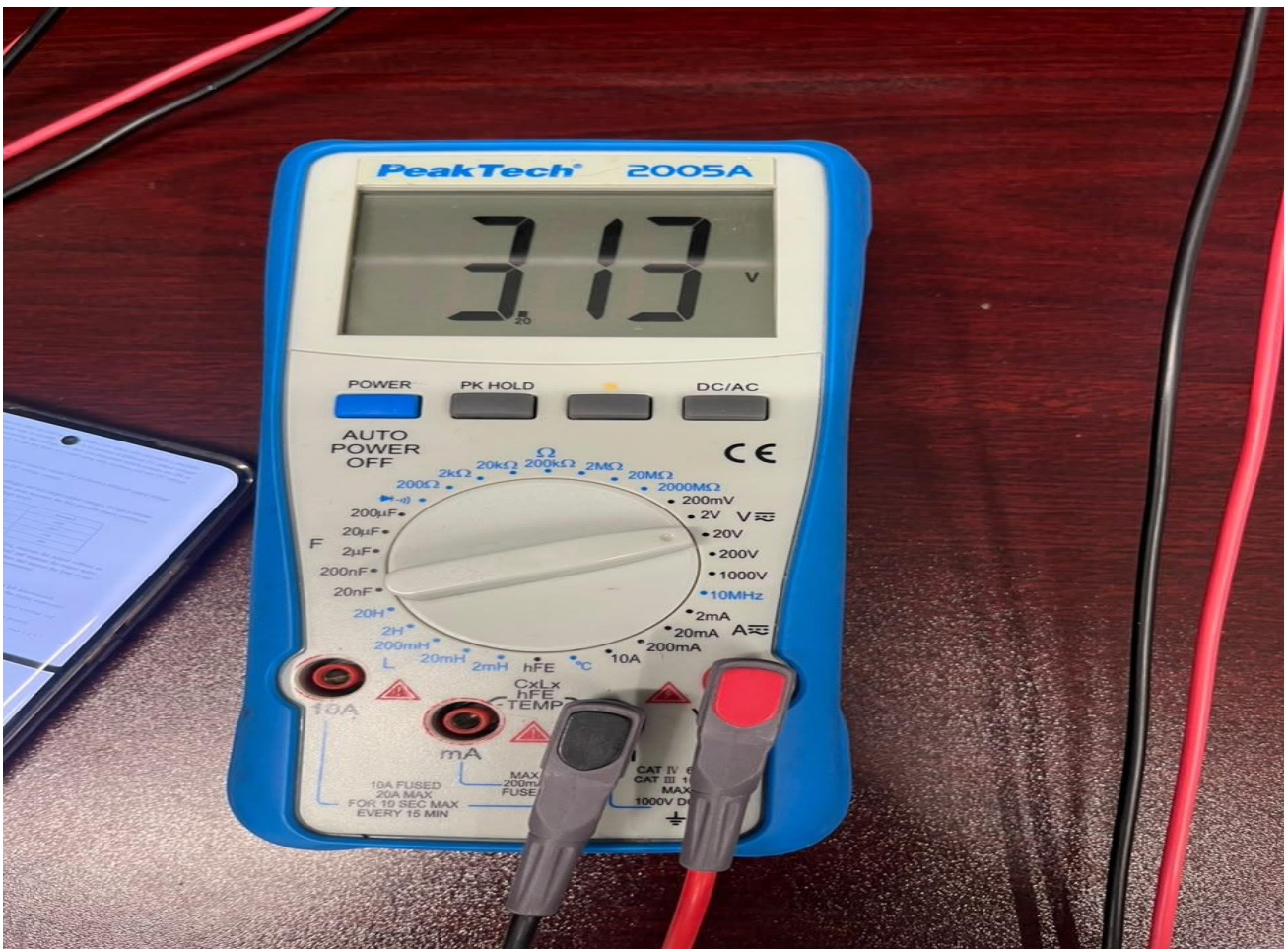
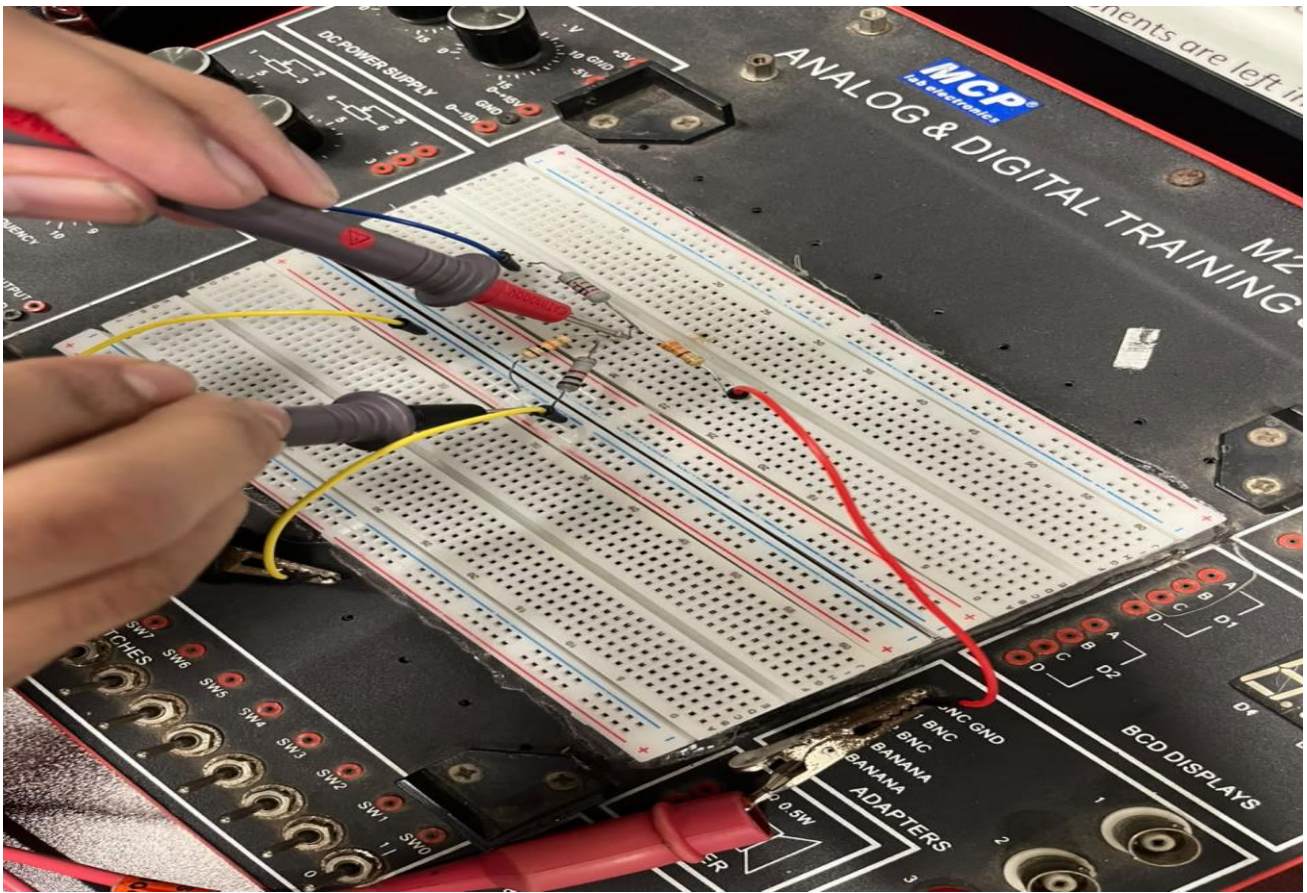
Component	Nominal value	Measured value	Deviation from nominal
R1	2.7 k Ω	2.74 k Ω	+1.5%
R2	3.3 k Ω	3.23 k Ω	−2.1%
R3	1.0 k Ω	1.00 k Ω	0.0%
RL	10 k Ω	9.57 k Ω	−4.3%

Measured values were obtained with the DMM in resistance mode before the resistors were installed in the powered circuit. Supply outputs were also measured directly: although both bench-supply displays read exactly 10 V and 6 V, the DMM measured $V_1 = 10.12$ V and $V_2 = 6.08$ V at the output terminals. All theoretical comparisons in this report use these DMM-measured supply voltages, not the display values, for consistency with the actual test conditions.

5.2 Breadboard Implementation

The final circuit was assembled on an 830-tie breadboard using the master topology. Both source negative terminals were tied to the same common ground rail, and R3 was returned to that same rail. Source branches were manually connected either to their supply voltage or to ground to represent ON and OFF.





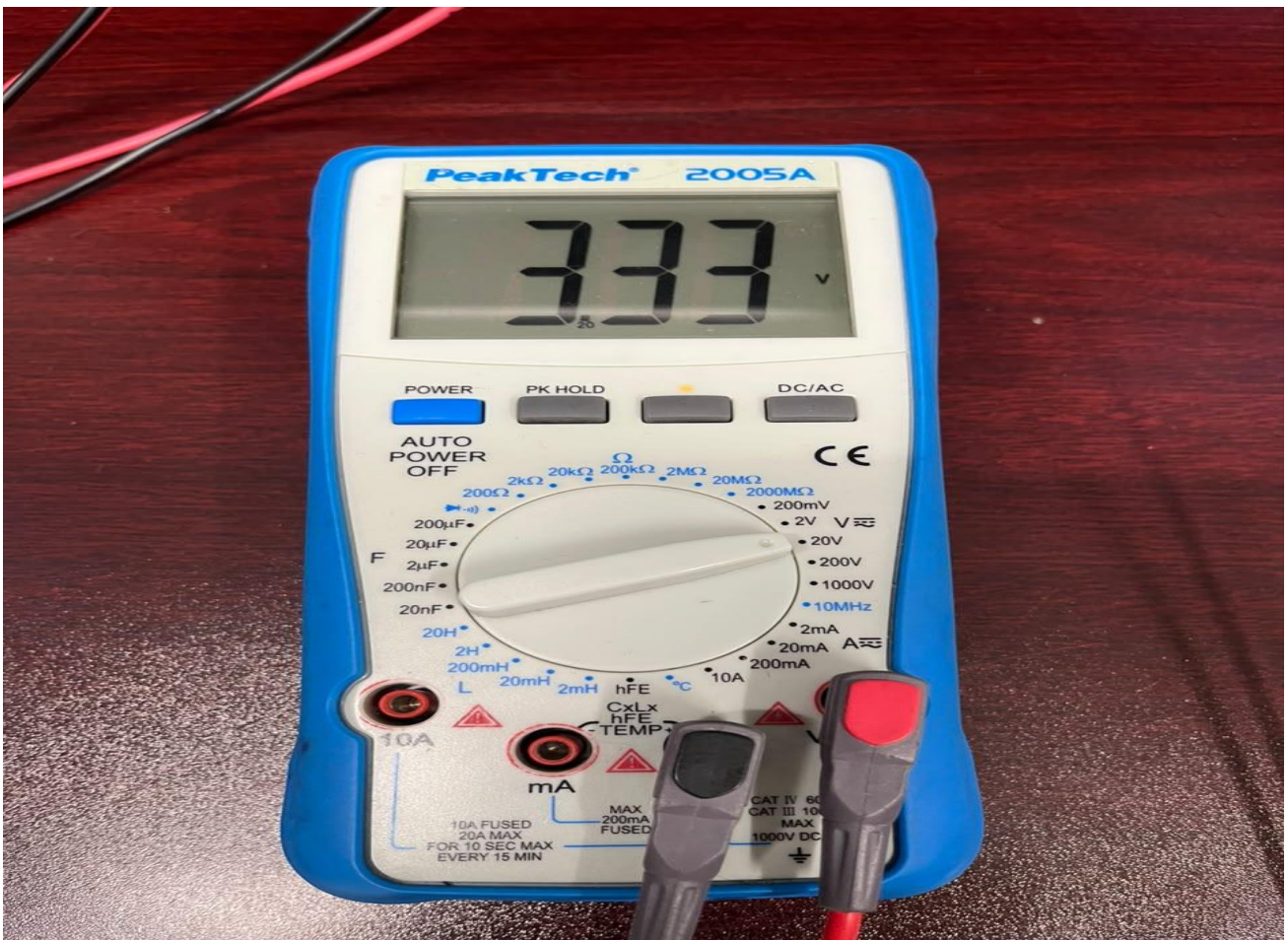
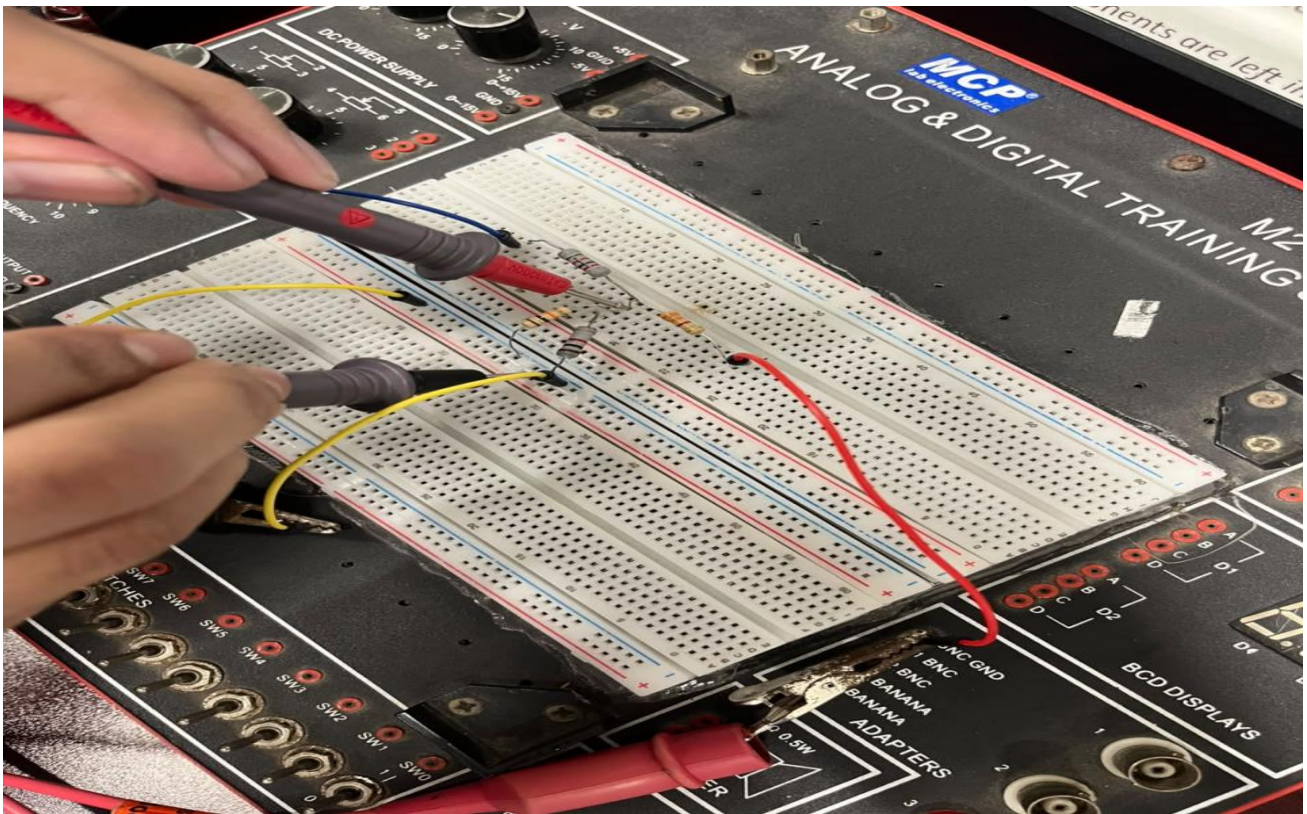


Figure: Final breadboard implementation of the two-source voltage monitoring circuit.

5.3 Hardware Testing Procedure

Node A was measured with the DMM in DC-voltage mode: red probe on Node A, black probe on the common ground rail. Each state was tested first without RL and then with the 10 k Ω monitoring resistor connected from Node A to ground.

5.4 Hardware Measurement Table

State	Measured V (no-load)	Measured V (loaded, 10 k Ω)	Notes
S0	0.00 V	0.00 V	as expected
S1	2.24 V	2.08 V	
S2	1.11 V	1.05 V	
S3	3.33 V	3.13 V	

5.5 Problems Found and Corrections

Two hardware issues were identified and corrected during testing. First, the two DC bench-supply outputs were initially connected using the wrong channel's terminals, which produced meaningless output readings at Node A; switching both source leads to the correct channel outputs (matching V1 = 10 V and V2 = 6 V) resolved the issue and produced sensible, repeatable readings.

Second, part of the resistor network was initially built across the breadboard's centre gap, which splits the main terminal strip into two electrically separate halves. Because no jumper bridged the two halves, the circuit was unintentionally left open. This was identified during testing and corrected by rebuilding the entire network on a single side of the gap, restoring full continuity.

In addition, although both bench-supply displays read exactly 10 V and 6 V, direct DMM measurement showed slightly higher output voltages (10.12 V and 6.08 V, Section 5.1). All calculations in this report were therefore based on the DMM-measured supply and resistor values rather than the nominal/displayed values.

6. Results and Discussion

6.1 No-Load Comparison

State	Target	Calculated	Simulated	Measured
S0	0 V	0.000 V	0.000 V	0.00 V
S1	2.2 V	2.213 V	2.213 V	2.24 V
S2	1.1 V	1.087 V	1.087 V	1.11 V
S3	—	3.300 V	3.300 V	3.33 V

6.2 Loaded Comparison

State	Calculated	Simulated	Measured
S0	0.000 V	0.000 V	0.00 V
S1	2.089 V	2.089 V	2.08 V
S2	1.025 V	1.025 V	1.05 V
S3	3.114 V	3.114 V	3.13 V

6.3 Specification Compliance

Spec	Requirement	Met?
1	S1 and S2 no-load results within $\pm 5\%$ of target	Y
2	S3 no-load result within $\pm 5\%$ of the calculated superposition value	Y
3	All adjacent loaded-state gaps ≥ 0.50 V	Y
4	Resistor/current/source/ground constraints satisfied	Y
5	Same nominal resistor values used in PSpice and hardware	Y

All five specifications are satisfied by the calculated, simulated, and measured results.

6.4 Discussion

The calculated and simulated results agree to within PSpice's own display rounding (under 0.001 V) at every state, both no-load and loaded — confirming the PSpice schematic is wired exactly as designed, with no simulation-side error.

The largest calculated-vs-measured difference is +0.030 V at S3 no-load (3.300 V calculated vs 3.33 V measured, +0.9%). This is fully explained by two independent, measured causes rather than a wiring fault: the actual resistor values ($R1 = 2.74$ k Ω , $R2 = 3.23$ k Ω , $R3 = 1.00$ k Ω , $R_L = 9.57$ k Ω) differ from nominal by up to -4.3% , and the actual supply outputs ($V1 = 10.12$ V, $V2 = 6.08$ V) sit about 1.2–1.3% above their nominal settings. Recalculating the expected output from these actual measured resistor and supply values predicts 3.330 V for S3 no-load — matching the 3.33 V measured value almost exactly — which confirms the small gap is component and supply tolerance, not a design or wiring error.

Adding the $10\text{ k}\Omega$ load reduces every output level (e.g. S3: $3.300\text{ V} \rightarrow 3.114\text{ V}$ calculated), because RL provides an additional path to ground in parallel with R3, lowering the equivalent resistance from Node A to ground and increasing the fraction of each source's voltage dropped across R1/R2 instead of appearing at the output. All four loaded levels remain clearly distinguishable, with adjacent gaps of roughly 1.0 V — more than double the 0.50 V minimum required by Spec 3.

7. Conclusion

This project successfully designed, simulated, and built a three-resistor summing-junction circuit ($R_1 = 2.7\text{ k}\Omega$, $R_2 = 3.3\text{ k}\Omega$, $R_3 = 1.0\text{ k}\Omega$ nominal) that produces four clearly distinguishable DC output voltages corresponding to the four ON/OFF combinations of the assigned 10 V and 6 V sources. All five design specifications were satisfied: the no-load S1 and S2 outputs fell within +1.82% and +0.91% of their assigned targets respectively; the no-load S3 output matched the calculated superposition value to within +0.92%; the loaded-state output levels remained separated by at least 1.0 V, well above the required 0.50 V minimum; branch currents and component values stayed within all specified limits; and identical nominal resistor values were used in both the PSpice simulation and the hardware build.

The calculated, simulated, and measured results agreed closely throughout, with the largest discrepancy (+0.9% at S3 no-load) fully explained by the measured tolerances of the actual resistors (up to -4.3% from nominal) and the small excess in the actual supply voltages over their nominal settings, rather than by any design or wiring fault. The key engineering decision in this design was the inclusion of a third resistor (R_3) from the output node to ground: with only two resistors, the single-source targets for S1 and S2 could not be satisfied simultaneously, since both states reduce to the same voltage-divider ratio; adding R_3 supplied the extra degree of freedom needed to meet both targets independently while also satisfying the loaded-state separation and current requirements.

One practical improvement for future iterations would be to verify the power-supply channel assignment and breadboard continuity with the DMM before wiring the full circuit, rather than after — both hardware issues encountered during this project (an incorrect supply channel and an unintended break across the breadboard's centre gap) would have been caught immediately by this simple check.

8. Appendix

8.1 Group Contribution Statement

Name	Contribution	ID
Ruhinul Haque Bhuiyan	Hardware construction and final resistor-combination selection	1000058247
Manajir Tasbik Murshed	Theoretical calculations and analysis	1000055271
Dibyendu Roy Dibya	PSpice simulation	1000058422

8.2 Component Cost Summary

Component	Quantity	Unit price (Tk)	Total (Tk)
Resistor set (2.7 k Ω , 3.3 k Ω , 1.0 k Ω , 10 k Ω)	1 set (4 pcs)	10	10
830-tie breadboard	1	120	120
Jumper wires	20	1	20
Grand Total			150

8.3 Supplementary Calculations

None required beyond Section 3.